

Cell-Driven Remodeling of Granular Hydrogel Scaffolds: A Discrete Element Model with Free Energy Landscape Analysis

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Significance

Granular hydrogel scaffolds—jammed packings of soft microparticles—are a promising platform for engineering tissues with controlled microarchitecture. When cell-laden (functional) granules are mixed with passive (inert) granules, cells generate traction forces that compact the functional phase, producing programmable shape transformations. We present a discrete element method (DEM) simulation that captures the essential physics of this process—Hertzian contact, motor-clutch cell traction, area-dependent friction, and superellipsoidal granule shapes—together with a mean-field compaction model and a free energy landscape framework that decomposes the competition between cell driving and mechanical resistance into six physical mechanisms. The resulting design rules map scaffold parameters to tissue architectures spanning bone, mucosa, kidney, and cardiac muscle.

Abstract

Granular hydrogel scaffolds composed of jammed hydrogel microparticles provide a versatile platform for tissue engineering, offering tunable porosity, mechanics, and biological function. When a fraction of the granules are laden with cells (functional) and the remainder are passive (inert), cell-generated traction forces drive selective compaction of the functional phase, producing macroscopic shape transformations. Here we develop GELLS-DEM, a two- and three-dimensional overdamped discrete element method that resolves individual granule contacts through Hertzian mechanics, motor-clutch cell traction, area-dependent hydrogel friction, DMT adhesion, and superellipsoidal particle shapes. We derive a volume-conserving two-zone mean-field model that reduces the many-body DEM dynamics to a single ordinary differential equation for the compaction coordinate ξ , and construct a free energy landscape $\tilde{G}(\xi)$ decomposed into six competing physical mechanisms: cell traction, Hertzian elastic resistance, Herschel-Bulkley yield stress near jamming, osmotic void redistribution, inert-granule geometric frustration, and functional-inert interfacial tension. Five dimensionless groups—the cell-elastic ratio β , cellular capillary number Ca , inert fraction Φ_r , packing proximity Ψ , and interfacial number Γ —govern the landscape topology and determine the equilibrium compaction ξ^* . A 5000-sample Latin hypercube parameter sweep maps the scaffold design space to tissue architectural descriptors, yielding quantitative design rules for achieving target organ microstructures including trabecular bone ($\xi^* \approx 0.58$), intestinal mucosa ($\xi^* \approx 0.38$), kidney cortex ($\xi^* \approx 0.21$), and cardiac muscle ($\xi^* \approx 0.10$).

Keywords: granular hydrogels | discrete element method | tissue engineering | energy landscape | cell mechanics

Author contributions: A.D.M. designed the study, developed the simulation code, performed the analysis, and wrote the paper.

Introduction

Granular hydrogels—dense packings of soft polymeric microparticles—have emerged as a versatile class of biomaterials for tissue repair and regeneration Riley et al. [2019]. Unlike monolithic hydrogels, granular scaffolds possess inherent porosity at the inter-particle scale, enabling nutrient transport, cell migration, and waste removal without the need for engineered pore networks Hollister [2005]. The mechanical properties of these scaffolds are governed by the Hertzian contacts between individual building blocks rather than by the bulk polymer network, providing an additional axis of tunability through particle stiffness, size, and shape Cai et al. [2022].

A particularly compelling application is the use of biphasic granular composites in which cell-laden “functional” granules are mixed with passive “inert” granules at a prescribed ratio Di Caprio et al. [2025]. Cells within the functional phase generate traction forces through the motor-clutch mechanism Chan and Odde [2008], Bangasser et al. [2013], forming intercellular bridges that compact the functional granules while the inert phase resists rearrangement. The resulting differential compaction produces macroscopic shape transformations whose geometry depends on the spatial distribution and volume fraction of each phase.

Despite the promise of this approach, rational design of granular scaffolds for specific tissue architectures remains largely empirical. The challenge is fundamentally multi-physics: cell-generated forces must overcome elastic resistance from Hertzian contacts, yield stress from the jammed packing O’Hern et al. [2003], van Hecke [2010], friction between granule surfaces Gong [2006], Pitenis et al. [2014], and geometric constraints from the inert phase. A predictive framework requires both a particle-resolved simulation that captures these interactions and a reduced-order model that identifies the governing

dimensionless groups.

Here we present GELLS-DEM (Granular Encapsulated Living-cell Laden Scaffold—Discrete Element Method), a simulation framework that addresses both needs. The DEM engine resolves the overdamped dynamics of $N \sim 500$ –1000 superellipsoidal granules with Hertzian contact mechanics, motor-clutch cell traction, area-dependent friction, and DMT adhesion. A mean-field compaction model reduces the many-body dynamics to a single ODE for the functional zone fraction x_f . Finally, a free energy landscape analysis decomposes the total free energy into six competing mechanisms, yielding five dimensionless groups that govern scaffold behavior.

Model Formulation

Discrete Element Method.

We model the granular scaffold as a collection of N particles in an overdamped regime appropriate for microscale dynamics in viscous media ($Re \ll 1$) Purcell [1977]. Each granule i has position $\mathbf{x}_i$, radius R_i (or semi-axes a_i, b_i, c_i and blockiness exponents $n_{1,i}, n_{2,i}$ for superellipsoids), and type $g_i \in \{0, 1\}$ distinguishing functional ($g_i = 0$) from inert ($g_i = 1$). The equation of motion is

$$\gamma_i \frac{d\mathbf{x}_i}{dt} = \sum_j \mathbf{F}_{ij}^{\text{contact}} + \mathbf{F}_i^{\text{cell}} + \mathbf{F}_i^{\text{wall}}, \quad (1)$$

where $\gamma_i = \gamma_0 R_i$ is the drag coefficient scaled by particle size. The contact force between particles i and j comprises three components: Hertzian repulsion, DMT adhesion, and area-dependent tangential friction. Integration is by forward Euler with a velocity cap to prevent numerical blowup.

Hertzian contact. For two elastic spheres with overlap $\delta_{ij} = R_i + R_j - |\mathbf{x}_i - \mathbf{x}_j|$, the normal repulsive force follows Hertz theory

Johnson [1985]:

$$F_{\text{Hertz}} = \frac{4}{3} E^* \sqrt{R^*} \delta^{3/2}, \quad (2)$$

where $E^* = E/[2(1 - \nu^2)]$ is the effective modulus and $R^* = R_i R_j / (R_i + R_j)$ is the reduced radius. For superellipsoidal particles, the common-normal method Delaney and Cleary [2010], Jaklič and Leonardis [2000] determines the contact point, penetration depth, and local curvature radii $R_{\text{loc},i}$, $R_{\text{loc},j}$, which replace the global radii in Eq. 2. A Newton-Raphson solver (typically 4–8 iterations) finds the contact parameters, with a fast-path bypass for sphere–sphere contacts.

DMT adhesion. Normal attractive forces follow the Derjaguin-Muller-Toporov model Derjaguin et al. [1975], $F_{\text{adh}} = 2\pi W R^*$, where the work of adhesion W depends on surface chemistry: bare–bare ($W = 0.5 \text{ mJ/m}^2$), bare–collagen ($W = 1.0 \text{ mJ/m}^2$), and collagen–collagen ($W = 2.0 \text{ mJ/m}^2$). These values are deliberately conservative to maintain the small-overlap assumption ($\delta/R < 0.1$).

Area-dependent friction. Hydrogel-on-hydrogel friction is fundamentally area-dependent rather than load-dependent Gong [2006], Pitenis et al. [2014], Uruena et al. [2018]: $F_{\text{fric}} = \tau_0 A_{\text{contact}} \tanh(v_t/v_{\text{ref}})$, where the interfacial shear stress τ_0 ranges from 50 Pa (bare gemini) to 2000 Pa (collagen–collagen) and $A_{\text{contact}} = \pi R^* \delta$ is the Hertzian contact area.

Motor-clutch cell traction. Cells on functional granules generate traction through the motor-clutch mechanism Chan and Odde [2008], Bangasser et al. [2013]. Each cell exerts a force

$$F_{\text{mc}} = F_{\text{stall}} \cdot \frac{k_{\text{sub}}}{k_{\text{sub}} + k_{\text{opt}}} \cdot \alpha_{\text{FA}} \cdot f_{\text{engage}}, \quad (3)$$

where $F_{\text{stall}} = n_{\text{motors}} \times F_{\text{motor}}$ is the total stall force, $k_{\text{sub}} = \pi E a_{\text{cell}} / (1 - \nu^2)$ is the substrate spring constant at the cell scale, $k_{\text{opt}} = n_{\text{clutches}} \times k_{\text{clutch}}$ is the optimal clutch stiffness, $\alpha_{\text{FA}} \in [0, 1]$ is focal adhesion maturity Geiger et al. [2009], and $f_{\text{engage}} = k_{\text{on}} / (k_{\text{on}} + k_{\text{off}})$ is the steady-state clutch engagement fraction. The net cell force on each granule is the vector sum of all attached cells directed toward the nearest functional neighbor.

Cell lifecycle. Cells are tracked individually through five states: ATTACHED $\rightarrow$ SPREADING $\rightarrow$ PROLIFERATING $\rightarrow$ BRIDGING $\rightarrow$ SENESCENT Engler et al. [2006]. Cells attach to collagen-coated functional granule surfaces, spread from 20 μm -diameter spheres to 5 μm -tall ellipsoids (volume-conserving), and form intercellular bridges between neighboring functional granules via a Poisson process with rate k_{bridge} . Bridges ramp force over a formation time, persist across timesteps, and transition to senescence after sustained load.

Superellipsoidal shapes. Granule shape is parameterized as superellipsoids ($|x/a|^{n_1} + |y/b|^{n_1} \rangle^{n_2/n_1} + |z/c|^{n_2} = 1$ Liu et al. [2025], Yuan et al. [2019], where semi-axes (a, b, c) control size and aspect ratio while blockiness exponents (n_1, n_2) interpolate between ellipsoids ($n = 2$) and rounded cuboids ($n > 2$). Rotational dynamics follow overdamped quaternion integration, with torques arising from off-center contact forces.

Mean-Field Compaction Model.

The N -body DEM dynamics can be coarse-grained to a single scalar: the functional zone fraction $x_f(t) = V_f / (V_f + V_i)$, where V_f and V_i are the volumes occupied by the functional and inert zones, respectively. We define the compaction coordinate $\xi = 1 - x_f/x_{f,0} \in [0, \xi_{\text{max}}]$, where $x_{f,0}$ is the initial zone fraction and ξ_{max} is set by the close-packing limit.

The total solid fraction $\phi_{\text{solid}} = \phi_f + \phi_i$ is conserved. As the functional zone compacts (ξ increases), its local solid fraction rises

while void fraction decreases; simultaneously, the inert zone dilates to accommodate the displaced void space. The volume-conserving constraint yields the local void fractions:

$$\phi_{v,f} = 1 - \frac{\phi_{f,0}}{x_{f,0}(1 - \xi)}, \quad \phi_{v,i} = 1 - \frac{\phi_{i,0}}{1 - x_{f,0}(1 - \xi)}. \quad (4)$$

The overdamped dynamics of ξ follow from a balance of the cell driving stress and mechanical resistance Kramers [1940], Doi and Edwards [1986]:

$$\eta_{\text{eff}} \frac{d\xi}{dt} = \underbrace{\sigma_{\text{cell}}(t)}_{\text{cell driving}} - \underbrace{\sigma_0 \max(0, \xi - \xi_c)^\alpha}_{\text{elastic resistance}}, \quad (5)$$

where $\sigma_{\text{cell}}(t) = \sigma_\infty \cdot f_{\text{bridge}}(t) \cdot m(t)$ incorporates the time-dependent bridge formation fraction and focal adhesion maturity, $\sigma_0 = 0.005 E^*$ is the elastic resistance prefactor (calibrated to the DEM Hertzian response), and ξ_c is the onset compaction above which elastic contacts form. The effective viscosity η_{eff} and exponent α are fitted to DEM simulations.

Free Energy Landscape.

To understand *why* a given set of scaffold parameters produces a particular compaction, we construct a free energy $G(\xi)$ whose minimum ξ^* gives the equilibrium compaction Wales [2003]. The landscape is decomposed into six physically distinct contributions:

(i) *Cell traction (driving):*

$$G_{\text{cell}}(\xi) = \sigma_{\text{cell}} \ln(1 - \xi), \quad (6)$$

the only term that decreases monotonically with ξ , providing the thermodynamic driving force for compaction. The logarithmic form arises from the volume-conserving geometry: $dx_f/d\xi = -x_{f,0}$, so the work done per unit compaction diverges as $\xi \rightarrow 1$.

(ii) *Hertzian elastic resistance:*

$$G_{\text{elastic}}(\xi) = \frac{\sigma_0}{2} \max(0, \xi - \xi_c)^2, \quad (7)$$

obtained by integrating the elastic resistance stress from Eq. 5. The parabolic form is consistent with Hertzian scaling in the small-overlap regime.

(iii) *Herschel-Bulkley yield stress:*

$$G_{\text{yield}}(\xi) = c_y \sigma_0 \max(0, \xi - \xi_J)^\Delta, \quad (8)$$

where ξ_J is the compaction coordinate corresponding to the jamming transition ($\phi_{f,\text{local}} = \phi_J$), $\Delta = 3/2$ is the O’Hern scaling exponent for Hertzian particles O’Hern et al. [2003], and $c_y = 0.25$ Olsson and Teitel [2007, 2012].

(iv) *Osmotic void redistribution:*

$$G_{\text{void}}(\xi) = \Pi [\Delta \phi_{v,f}^2 x_f + \Delta \phi_{v,i}^2 x_i], \quad (9)$$

where $\Pi = 3\sigma_{\text{cell}}$ is the osmotic pressure scale and $\Delta \phi_v$ measures the deviation of local void fraction from its initial value in each zone Biot [1941], Princen [1986]. This term penalizes the asymmetric redistribution of void space during compaction.

(v) *Inert geometric frustration:*

$$G_{\text{inert}}(\xi) = k_f \frac{\phi_i}{\phi_f} \xi^2, \quad (10)$$

where $k_f = 1.5\sigma_{\text{cell}}$ captures the energy cost of rearranging functional granules around inert obstacles Bi et al. [2014, 2015]. The ϕ_i/ϕ_f prefactor ensures that this term vanishes in the absence of inert granules.

(vi) *Interfacial tension*:

$$G_\gamma(\xi) = \gamma[\sqrt{1-\xi} - 1] + \gamma\phi_i\xi/2, \quad (11)$$

where γ depends on the modulus mismatch between functional and inert phases Princen [1979, 1983]. The first term represents the contraction of the functional–inert boundary as the functional zone compacts; the second penalizes forced mixing of inert granules into the functional domain.

The total normalized landscape is $\tilde{G}(\xi) = G(\xi)/\sigma_{\text{cell}}$, giving $O(1)$ values that facilitate comparison across parameter regimes. The equilibrium ξ^* satisfies $d\tilde{G}/d\xi = 0$, $d^2\tilde{G}/d\xi^2 > 0$.

Dimensionless Groups.

Five dimensionless numbers emerge from the energy landscape and govern the compaction behavior:

$$\beta = \frac{\sigma_{\text{cell}}}{\sigma_0} = \frac{\text{cell traction}}{\text{elastic resistance}}, \quad (12)$$

$$\text{Ca} = \frac{\sigma_{\text{cell}}}{\sigma_y} = \frac{\text{cell traction}}{\text{yield stress}}, \quad (13)$$

$$\Phi_r = \frac{\phi_i}{\phi_f} = \text{inert-to-functional ratio}, \quad (14)$$

$$\Psi = \frac{\phi_{\text{solid}}}{\phi_{\text{RCP}}} = \text{proximity to jamming}, \quad (15)$$

$$\Gamma = \frac{\gamma}{\sigma_{\text{cell}}R} = \frac{\text{interfacial tension}}{\text{cell stress} \times \text{size}}. \quad (16)$$

Large β and Ca favor deep compaction (cells easily overcome elastic and yield resistance); large Φ_r and Γ frustrate compaction; and Ψ close to unity places the system near the jamming point where yield stress diverges.

Results

DEM simulations of scaffold remodeling.

We performed a 23-run design of experiments (DOE) varying five factors: granule modulus E (2–50 kPa), cells per granule (4–16), functional fraction ϕ_f (0.2–0.5), granule radius R (20–80 μm), and simulation duration (24–72 h). The DEM captures the full microstructural evolution: initial random packing, cell attachment and spreading, bridge formation, selective compaction of the functional phase, and eventual kinetic arrest (Fig. 1).

Figure 1 shows representative timelapse sequences for four organ-targeted parameter sets. Trabecular bone conditions (low ϕ_f , soft granules) exhibit the most dramatic compaction ($\xi^* \approx 0.58$), with the functional phase condensing into a dense interconnected network surrounded by void-rich inert regions. Intestinal mucosa ($\xi^* \approx 0.38$) shows moderate compaction with preserved porosity. Kidney cortex ($\xi^* \approx 0.21$) and cardiac muscle ($\xi^* \approx 0.10$) represent progressively stiffer scaffolds where elastic resistance limits compaction.

Mean-field model captures compaction kinetics.

The mean-field ODE (Eq. 5) was fitted to each DOE run by optimizing η_{eff} and σ_0 against the DEM-computed $x_f(t)$ trajectory. Figure 2 shows the compaction kinetics for the four organ targets. The three-panel view reveals the coordinated dynamics: zone fraction $x_f(t)$ decreases as functional granules compact (left), void space is redistributed from the functional to the inert zone (center), and bridge formation follows a sigmoidal time course (right).

The mean-field model captures the essential features: a lag phase during cell attachment, rapid compaction during bridge formation, and saturation as elastic resistance balances cell traction. The best fits achieve $R^2 > 0.6$ for runs with substantial compaction ($\Delta\phi_f > 0.05$). Runs with minimal compaction ($E = 2$ kPa, few cells) show poor fits because the signal-to-noise ratio is low.

Energy landscape reveals competing mechanisms.

The free energy landscape $\tilde{G}(\xi)$ (Fig. 3) reveals how the six energy terms compete to determine equilibrium compaction. For each organ target, the landscape exhibits a single minimum at ξ^* whose position and depth depend on the relative magnitudes of driving and resistance terms.

For trabecular bone, the cell traction term (green) dominates, producing a deep minimum at $\xi^* = 0.58$. The elastic (blue) and yield (red) terms contribute modestly because the soft granules ($E = 3$ kPa) generate weak Hertzian resistance. For cardiac muscle, conversely, the stiff granules ($E = 20$ kPa) produce strong elastic resistance that arrests compaction at $\xi^* = 0.10$ despite active cell traction.

The time evolution of the landscape (Fig. 4) demonstrates how bridge formation progressively deepens the cell traction well, pulling the system from $\xi = 0$ toward ξ^* . Before bridge formation ($t < 6$ h), the landscape is flat and the system is uncompacted. As bridges form (6–18 h), the minimum deepens and shifts to larger ξ , driving rapid compaction. By $t = 36$ h, the landscape has reached its asymptotic form and compaction saturates.

Energy decomposition quantifies design trade-offs.

Figure 5 decomposes the energy balance at equilibrium for each organ target, alongside the five governing dimensionless groups. The left panel shows stacked energy contributions at ξ^* : negative bars represent driving terms (cell traction, interfacial contraction) and positive bars represent resistance (elastic, yield, void, frustration).

Trabecular bone achieves the deepest compaction ($\xi^* = 0.37$) because cell traction substantially exceeds all resistance terms combined. The dimensionless groups (right panel) confirm this: $\beta \gg 1$ (cells dominate elasticity), low Φ_r (few inert obstacles). Cardiac muscle is the opposite limit: $\beta \ll 1$, large Φ_r , yielding minimal compaction.

The osmotic void term (purple) becomes important at intermediate compaction, acting as a restoring force that penalizes asymmetric void redistribution. The interfacial tension term (orange) provides a modest additional resistance proportional to the modulus mismatch between functional and inert granules.

Design space maps scaffold parameters to tissue architectures.

A 5000-sample Latin hypercube sweep maps how scaffold inputs control the final void fraction (Fig. 6). Four projections reveal the principal design axes: (i) softer granules and lower ϕ_f produce the highest porosity ($\phi_v > 0.7$), suitable for trabecular bone and lung; (ii) the constraint $\phi_f + \phi_i < \phi_{\text{RCP}}$ bounds the accessible design space, with the organ-relevant region at $\phi_f \in [0.1, 0.4]$, $\phi_i \in [0.1, 0.3]$; (iii) granule radii control absolute pore size and Kozeny–Carman permeability at fixed composition; and (iv) modulus mismatch between phases introduces interfacial stress concentrations that modify compaction profiles.

Discussion

Three key insights emerge from the GELLS-DEM framework. First, the cell–elastic ratio $\beta = \sigma_{\text{cell}}/\sigma_0$ is the primary design knob. Al-

though $\sigma_{\text{cell}} \sim 10^{-3}$ kPa is small compared to the bulk modulus E , the relevant resistance is the Hertzian *contact* stress ($\sigma_0 = 0.005E^*$), yielding $\beta \sim 0.1$ –1 and explaining why soft hydrogels are essential for cell-driven remodeling. Second, the jamming transition imposes a hard ceiling: as $\phi_{f,\text{local}} \rightarrow \phi_J$, the yield stress diverges as $(\phi - \phi_J)^\Delta$ O’Hern et al. [2003], Olsson and Teitel [2012], so scaffolds near close packing ($\Psi > 0.95$) cannot compact further regardless of cell number. Third, inert granules provide tunable geometric frustration ($G_{\text{inert}} \propto \Phi_r \xi^2$): increasing the inert fraction shifts ξ^* without changing granule mechanics, enabling access to diverse tissue architectures with the same material system.

The landscape framework offers physical interpretability (diagnosing *why* a design fails), rational scaling (preserving β , Ca , Φ_r across granule sizes), and computational speed (ξ^* computed in milliseconds vs. hours for full DEM). Limitations include the spatially homogeneous mean-field assumption Tighe et al. [2010], steady-state motor-clutch approximation, DMT adhesion ($\sim 25\%$ overestimate vs. JKR, compensated by conservative W values), and neglect of cell proliferation and ECM deposition beyond 72 h.

Materials and Methods

DEM implementation.

GELLS-DEM is implemented in Python 3.9+ (NumPy, SciPy). Neighbor search uses `ckdTree` with cutoff $2R_{\text{max}} + L_{\text{max}}$; optional Numba JIT accelerates the Newton-Raphson solver. Domain: rectangular box with rigid walls. Integration: forward Euler, $v_{\text{max}} = 50 \mu\text{m/h}$. Initial packings by random sequential addition plus 200-step compression settle; radii from log-normal distributions; types assigned randomly at target ratio.

Parameter sweep and tissue targets.

A 5000-sample Latin hypercube spans $\log_{10} E_{\text{func}}$ (0–2.3), ϕ_f (0.05–0.55), ϕ_i (0.05–0.55), $\log_{10} R_{\text{func}}$ (1.2–1.9), $\log_{10} R_{\text{inert}}$ (1.2–1.9), subject to $\phi_f + \phi_i \geq \phi_{\text{RCP}}$. Seven organ targets (bone, mucosa, kidney, cardiac, liver, lung, pancreas) are defined from literature BV/TV, Tb.Th, Tb.Sp, Conn.D, DA, ϕ_v , and κ values Hollister [2005], Harrigan and Mann [1984], Hildebrand and Rüegsegger [1997]. The energy landscape is evaluated and ξ^* computed by root-finding for each sample; tissue descriptors are mapped to targets via weighted Euclidean distance. Source code: <https://github.com/McGheeLab/GELLS-DEM>.

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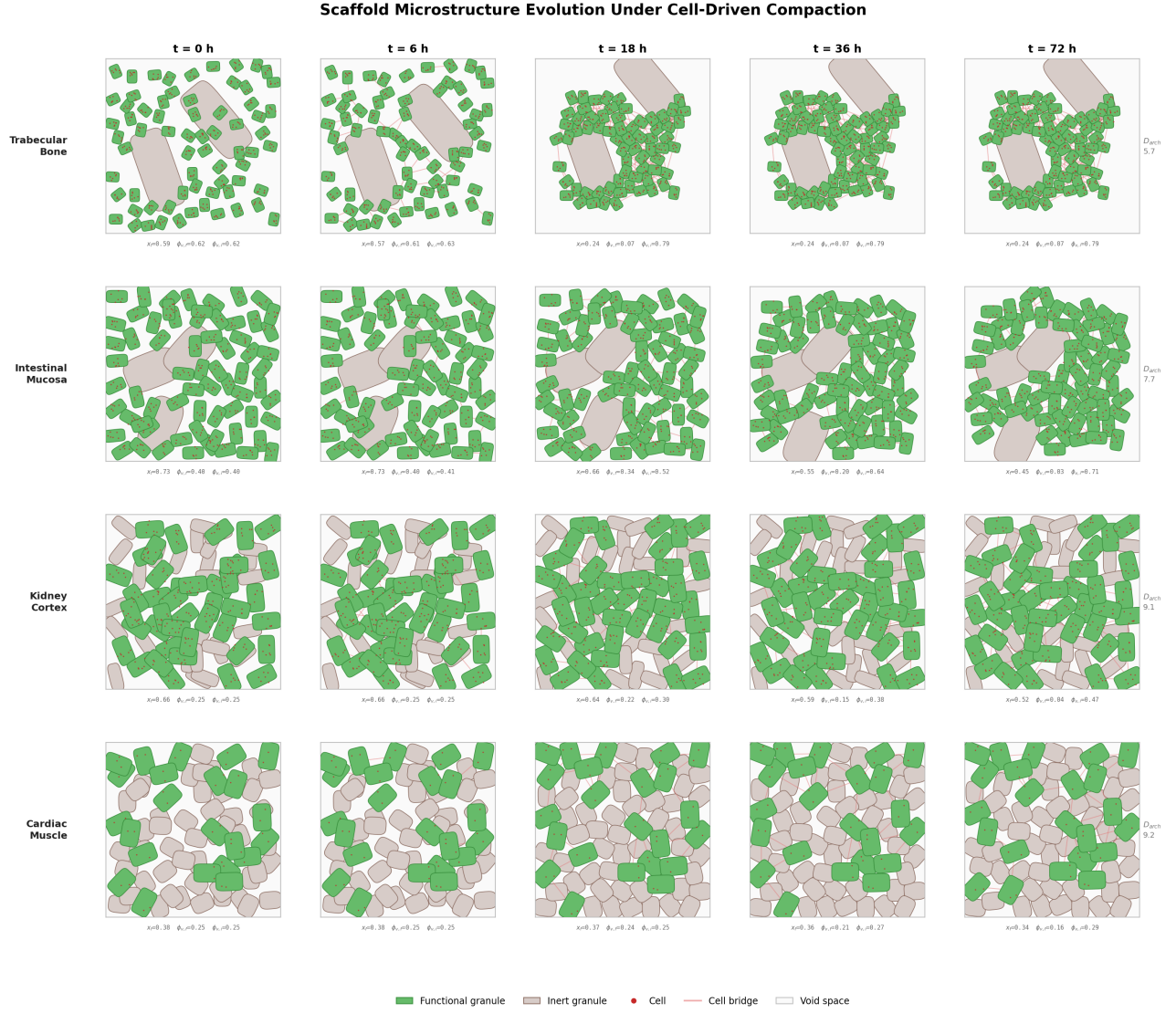


Figure 1. Scaffold microstructure evolution under cell-driven compaction. Timelapse sequences at $t = 0, 6, 18, 36,$ and 72 h for four organ targets: trabecular bone (row 1), intestinal mucosa (row 2), kidney cortex (row 3), and cardiac muscle (row 4). Functional (cell-laden) granules shown in green; inert in gray. Superellipsoidal shapes with varying aspect ratio and blockiness are visible. The functional phase selectively compacts via intercellular bridges while the inert phase remains stationary. Compaction increases from cardiac muscle (minimal) to trabecular bone (extensive). Inset bar charts show evolving volume fractions (ϕ_f, ϕ_i, ϕ_v), confirming solid conservation and void redistribution.

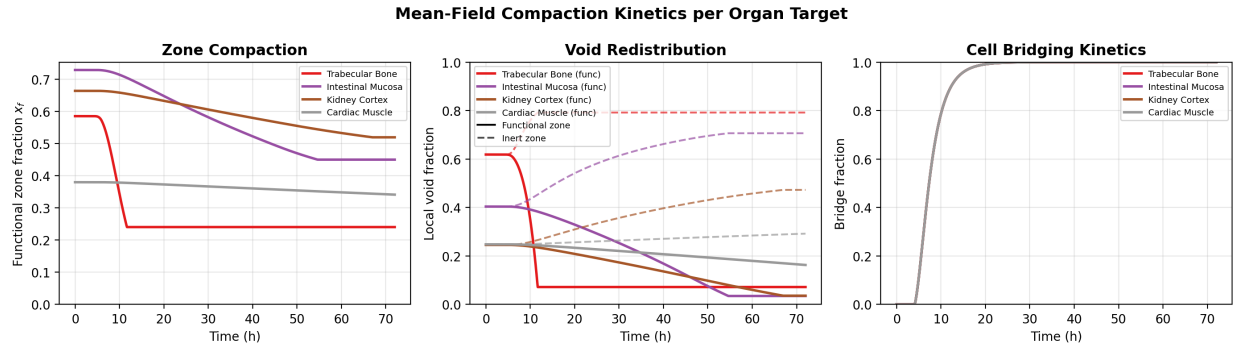


Figure 2. Mean-field compaction kinetics per organ target. **Left:** Functional zone fraction $x_f(t)$. Trabecular bone (red) compacts from $x_f = 0.65$ to 0.28 ; cardiac muscle (gray) shows minimal change. **Center:** Local void fractions in functional (solid) and inert (dashed) zones. Compaction drives functional void fraction toward zero while inert zone dilates. **Right:** Bridge fraction $f_{\text{bridge}}(t)$ with sigmoidal onset at ~ 6 h (post-attachment) and saturation by ~ 18 h.

Free Energy Landscape for Cell-Driven Scaffold Compaction

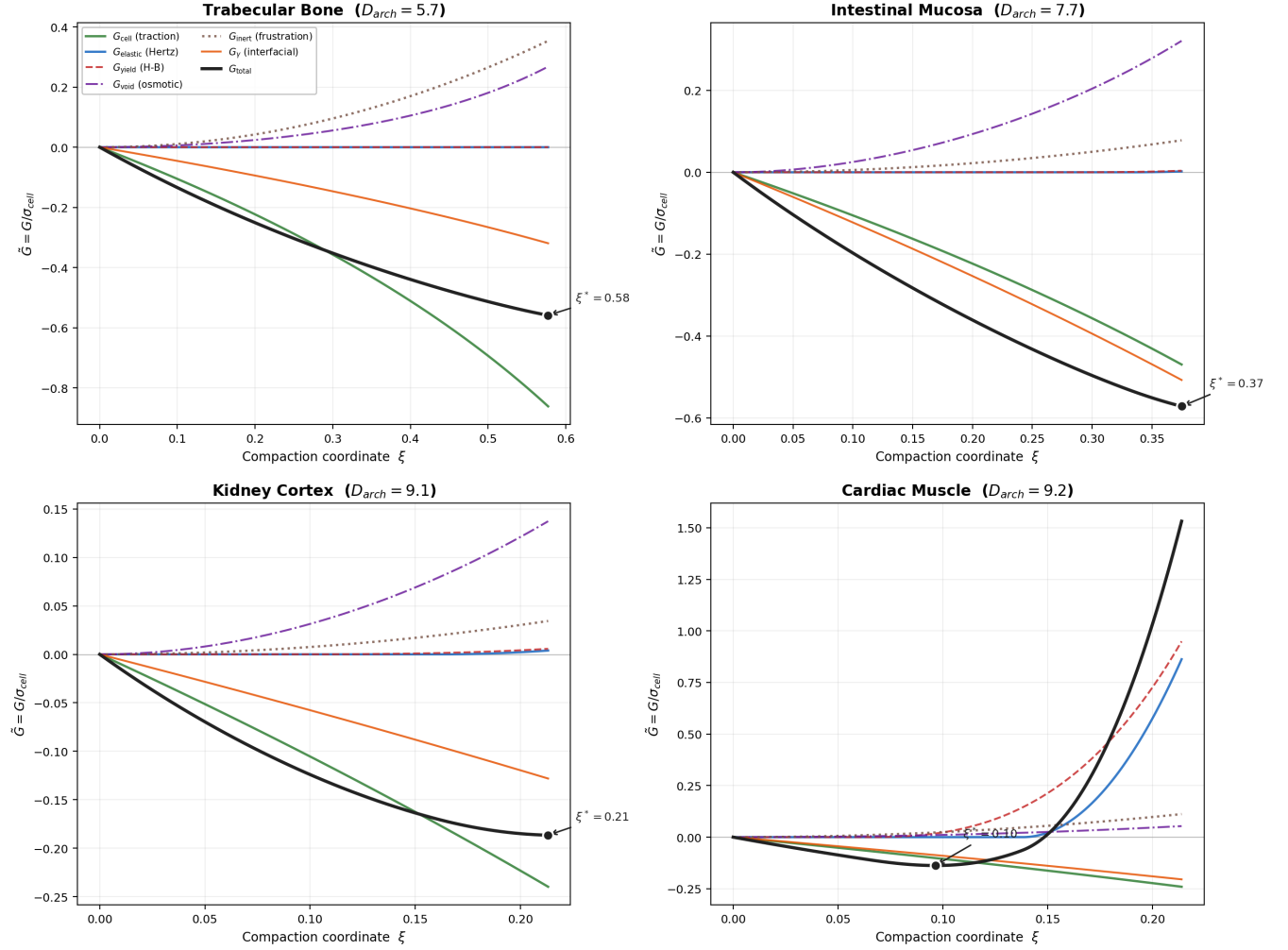


Figure 3. Free energy landscape for cell-driven scaffold compaction. Normalized $\tilde{G}(\xi) = G(\xi)/\sigma_{\text{cell}}$ for four organ targets. Six contributions: G_{cell} (green, driving), G_{elastic} (blue), G_{yield} (red dashed), G_{void} (purple), G_{inert} (brown), G_{γ} (orange). Total landscape (black) has a single minimum at ξ^* (filled circle). Trabecular bone: $\xi^* = 0.58$, deep minimum (cell traction dominates). Cardiac muscle: $\xi^* \approx 0.10$, shallow minimum (strong elastic resistance). D_{arch} denotes architectural distance to target.

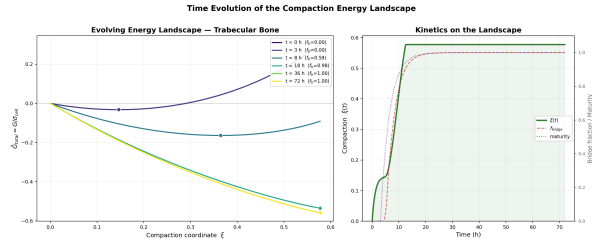


Figure 4. Time evolution of the compaction energy landscape. **Left:** $\tilde{G}(\xi)$ for trabecular bone at six timepoints. Bridge formation (f_b) progressively deepens the cell traction well; filled circles mark instantaneous $\xi(t)$ tracking the evolving minimum. **Right:** $\xi(t)$ (black), $f_{\text{bridge}}(t)$ (red dashed), and FA maturity (gray dotted). Compaction onset lags bridge formation by ~ 2 h due to viscous damping.

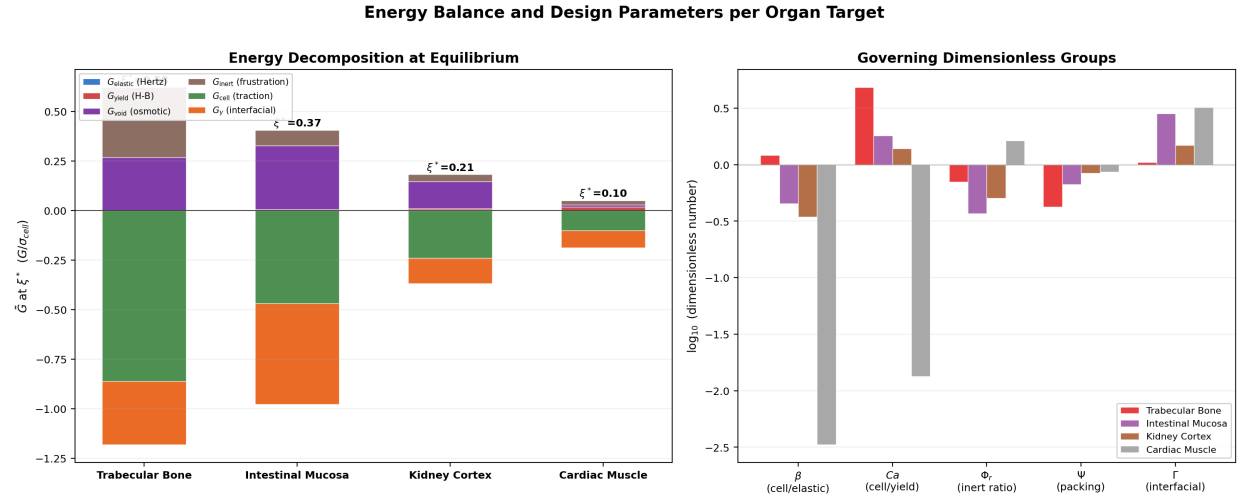


Figure 5. Energy decomposition and governing dimensionless groups. **Left:** Stacked energy at ξ^* for four organ targets. Negative: driving (green: G_{cell} ; orange: G_{γ}). Positive: resistance (blue: elastic; red: yield; purple: void; brown: frustration). **Right:** Five dimensionless groups on $\log_{10}$ scale. Large β and Ca correlate with deep compaction; large Φ_r , Ψ , Γ resist it.

Final Void Fraction: 2-D Binned Heatmaps

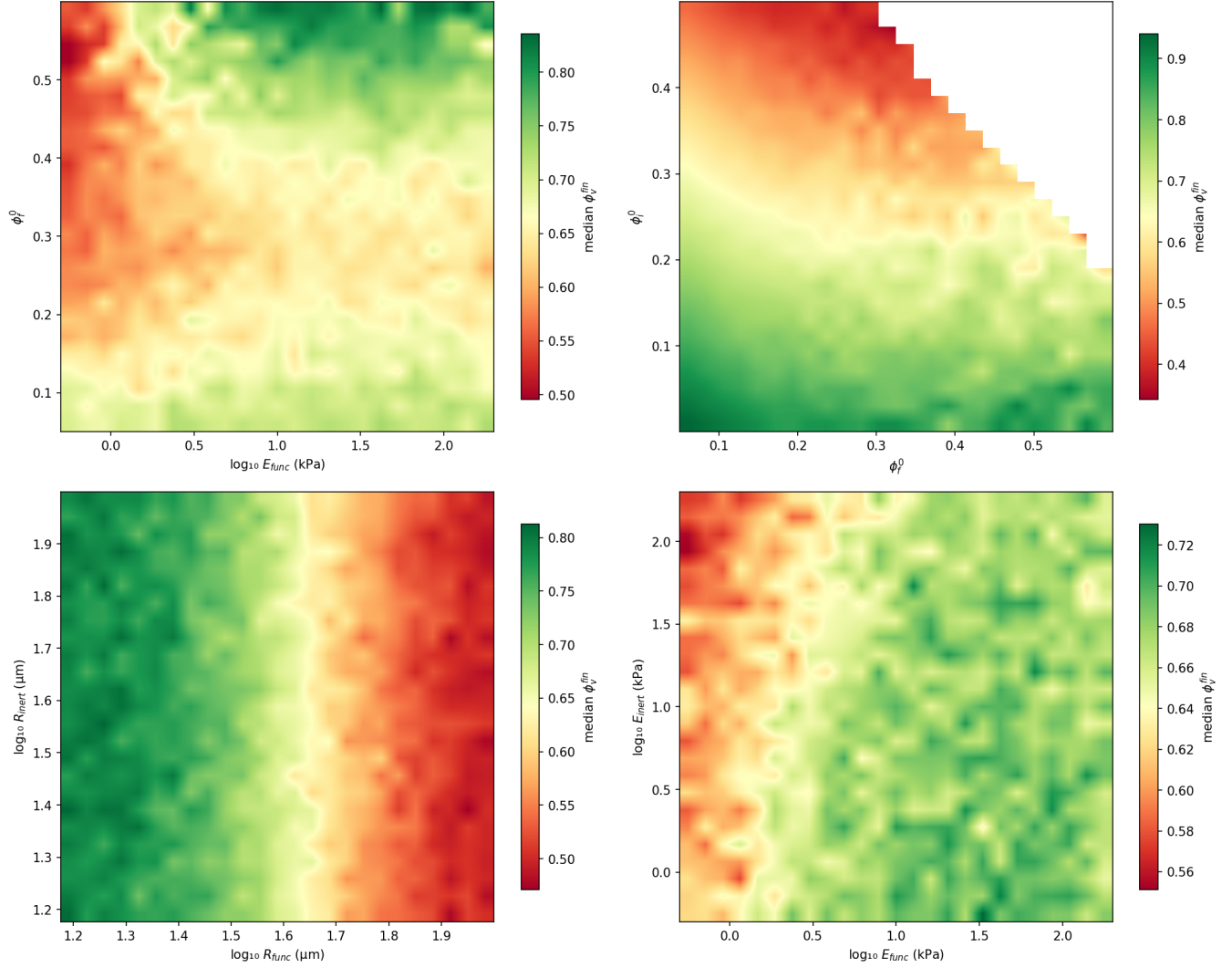


Figure 6. Design space for scaffold void fraction. Median ϕ_{vf}^* from 5000-sample Latin hypercube sweep. **Upper left:** Stiffness vs. ϕ_i ; soft granules yield high porosity. **Upper right:** ϕ_f vs. ϕ_i ; diagonal boundary from $\phi_f + \phi_i \leq 1$. **Lower left:** Granule radii control pore size and permeability. **Lower right:** Modulus mismatch modifies compaction via interfacial stress.